

DuET: Dual Incremental Object Detection via Exemplar-Free Task Arithmetic

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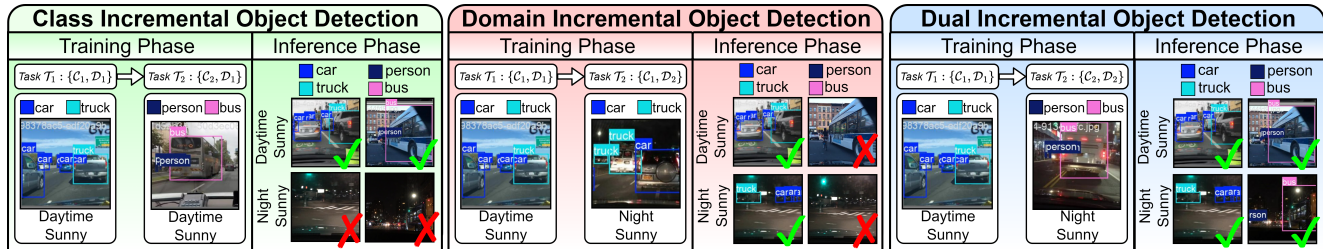


Figure 1. Comparison of Class Incremental Object Detection (CIOD), Domain Incremental Object Detection (DIOD), and our proposed Dual Incremental Object Detection (DuIOD), where $\checkmark$: the scenario is addressed, $\times$: the scenario is NOT addressed. **CIOD (left)** incrementally learns new categories ($C_1 \rightarrow C_2$) in a fixed domain but struggles in unseen domains. **DIOD (middle)**, adapts to new domains ($D_1 \rightarrow D_2$) but does not detect new classes. In contrast, **DuIOD (right)** addresses both challenges by learning new object categories while generalizing across evolving domains (zoomed-in for best view).

Abstract

Real-world object detection systems, such as those in autonomous driving and surveillance, must continuously learn new object categories and simultaneously adapt to changing environmental conditions. Existing approaches, Class Incremental Object Detection (CIOD) and Domain Incremental Object Detection (DIOD)—only address one aspect of this challenge. CIOD struggles in unseen domains, while DIOD suffers from catastrophic forgetting when learning new classes, limiting their real-world applicability. To overcome these limitations, we introduce Dual Incremental Object Detection (DuIOD), a more practical setting that simultaneously handles class and domain shifts in an exemplar-free manner. We propose DuET, a Task Arithmetic-based model merging framework that enables stable incremental learning while mitigating sign conflicts through a novel Directional Consistency Loss. Unlike prior methods, DuET is detector-agnostic, allowing models like YOLO11 and RT-DETR to function as real-time incremental object detectors. To comprehensively evaluate both retention and adaptation, we introduce the Retention-Adaptability Index (RAI), which combines the Average Retention Index (Avg RI) for catastrophic forgetting and the Average Generalization Index for domain adaptability into a common ground. Extensive experiments on the Pascal Series and Diverse Weather Series demonstrate DuET’s effectiveness, achieving a +13.12% RAI improvement while preserving 89.3% Avg RI on the Pascal Series (4 tasks), as well as a +11.39% RAI improvement with 88.57% Avg RI on the Diverse Weather Series (3 tasks), outperforming existing methods.

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1. Introduction

Real-world object detection systems must continuously learn new categories and adapt dynamically to evolving environments without necessitating full retraining. Traditional models, typically trained on static datasets, struggle to perform effectively in dynamic real-world scenarios like autonomous driving and surveillance, where they must recognize novel objects while handling environmental variations. For example, an autonomous vehicle must detect newly introduced road signs under diverse lighting and weather conditions—without forgetting past knowledge, a challenge known as catastrophic forgetting [25, 38, 40].

Existing approaches attempt to address this through Class Incremental Object Detection (CIOD) [9, 19, 20, 28, 40], which learns new object categories while assuming a fixed domain, and Domain Incremental Object Detection (DIOD) [41], which adapts to unseen domains while keeping object categories unchanged. However, these methods operate under restrictive assumptions—CIOD neglects domain shifts, while DIOD assumes a fixed set of object categories. As shown in Figure 1, both approaches fail when faced with simultaneous class and domain shifts, a scenario frequently encountered in real-world applications.

To bridge this gap, we introduce a more practical yet underexplored setting, Dual Incremental Object Detection (DuIOD), where object detectors must incrementally learn new object categories across evolving domains without access to previous training data (exemplar-free). While some works, such as VIL [35], extend incremental learning to both class and domain shifts, they primarily focus on classification tasks and do not address key object detection chal-

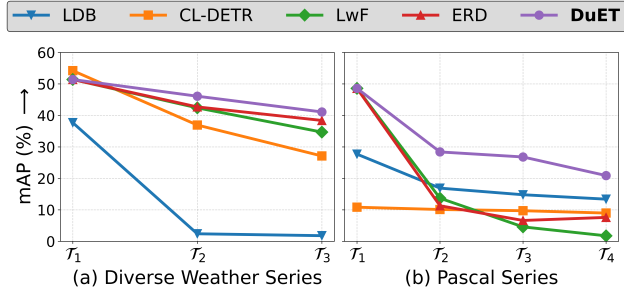


Figure 2. **Catastrophic forgetting of T_1 classes in multi-phase experiments for various methods on the DuIOD task.** The *left plot* illustrates catastrophic forgetting in *Night Sunny* [1:2] classes across three incremental tasks: Night Sunny [1:2] (T_1) → Daytime Sunny [3:4] (T_2) → Daytime Foggy [5:7] (T_3) on the Diverse Weather Series dataset. Similarly, the *right plot* depicts catastrophic forgetting in *Watercolor* [1:3] classes across four incremental tasks: Watercolor [1:3] (T_1) → Comic [4:6] (T_2) → Clipart [7:13] (T_3) → VOC [14:20] (T_4) on the *Pascal Series* dataset.

allenges like spatial feature retention and bounding box refinement, nor do they tackle the background shift problem of IOD [27, 32], where previously learned objects are misclassified as background if unannotated in later tasks, all of which are present in DuIOD. Recently, PD [15] partially addresses domain shifts in object detection but fails to generalize to the broader DuIOD problem.

To tackle the challenges of DuIOD, we propose DuET, a simple yet effective Task Arithmetic (TA) [13] based incremental learning framework for robust object detection across evolving domains. TA paradigm was recently introduced by [13] for modifying pre-trained models via arithmetic operations on task vectors. Formally, a task vector is derived by subtracting the weights of a pre-trained model from those of the same model fine-tuned on a specific task. In DuET, we decompose model parameters into shared parameters, which persist across all incremental tasks, and task-specific parameters, which evolve incrementally. We compute task vectors for both previous and current tasks using the shared parameters. DuET framework is built upon two components: *DuET Module & Incremental Head*. *DuET Module* merges these task vectors through dynamic layer-wise retention and adaptation factors, effectively mitigating catastrophic forgetting. Meanwhile, the *Incremental Head* concatenates task-specific parameters from both past and current tasks, enhancing generalization. Unlike prior methods, DuET is object detector-agnostic, seamlessly integrating with diverse detection backbones. Notably, it enables state-of-the-art real-time object detectors, such as YOLO11 [17] and RT-DETR [29], to function as real-time incremental object detectors.

Additionally, we introduced Directional Consistency Loss to further stabilize incremental updates, which mitigates weight update conflicts during model merging [45, 49]. Moreover, to jointly quantify catastrophic forgetting

and adaptation performance, we propose a new evaluation metric, the Retention-Adaptability Index (RAI).

Our extensive experiments on *Pascal Series* and *Diverse Weather Series* datasets demonstrate the effectiveness of DuET in addressing the DuIOD task. As highlighted in Figure 2, CIOD-based methods (e.g., CL-DETR [28]) suffer severe forgetting in unseen domains, while DIOD-based methods (e.g., LDB [41]) fail to retain previously learned categories. Additionally, conventional incremental learning techniques such as LwF [25] and ERD [8] also struggle in this setting. In contrast, DuET significantly mitigates catastrophic forgetting and enables robust incremental learning across both class and domain shifts. We highlight our contributions as:

- To the best of our knowledge, DuET is the first method to tackle DuIOD using the concept of Task Merging.
- We propose a novel TA-based algorithm which is object detector agnostic, validated YOLO11 and RT-DETR, enabling real-time incremental object detection.
- We propose Directional Consistency Loss to mitigate sign conflicts during model merging, ensuring stable incremental learning.
- To quantify both catastrophic forgetting and adaptation performance, we introduce a new evaluation metric, the Retention-Adaptability Index.

2. Related Works

Class Incremental Object Detection: CIOD has evolved from early CNN-based methods [2, 8, 22, 27, 34, 36, 37, 40] to Transformer-based approaches [9, 18–20, 28], improving generalization and stability during incremental updates. Initial efforts, such as [40], introduced CIOD via knowledge distillation inspired by incremental classification methods like LwF [25]. Subsequent works explored intermediate feature distillation [3, 4, 10], replay buffers [1, 19, 39], parameter isolation [23, 50], and pseudo-labeling [9, 20, 46].

Transformer-based models, including CL-DETR [28], SDDGR [19], and VLM-PL [20], use *Deformable DETR* [52] to enhance CIOD. CL-DETR applies knowledge distillation on labels with exemplar replay, SDDGR integrates generative replay via stable diffusion, and VLM-PL employs vision-language pseudo-labeling. However, most CIOD methods rely on exemplars [1, 19, 51] and assume a static domain. In contrast, DuET proposes an exemplar-free, unified approach that prevents *catastrophic forgetting* while incrementally learning new objects across domains.

Domain Incremental Object Detection: DIOD focuses on adapting to distribution shifts—such as changes in lighting, viewpoints, and weather—while maintaining a fixed object set. LDB [41] tackles DIOD by learning domain biases with frozen base models and employing a nearest-mean classifier for domain selection, while other methods [33, 42, 43] primarily target *Domain Incremental Learn-*

ing through prompt-based strategies, domain statistics, and cross-domain concept alignment.

Although some DIOD methods support *exemplar-free adaptation*, they assume a consistent class set across domains and struggle with *inter-domain class confusion*. DuET overcomes these limitations by using **task arithmetic-based model merging** to incrementally learn class and domain shifts without relying on exemplars.

Task Arithmetic: Task arithmetic has recently emerged as an effective way to modify pre-trained models through simple arithmetic operations on model weights. Task-Arithmetic [13] showed that *task vectors*—differences between fine-tuned and pre-trained model weights—can steer model behaviour. Extensions like Fisher-Merging [31] and RegMean [16] introduced importance-weighted schemes using Fisher information and inner-product matrices, but they require manual matrix computations. To resolve *sign conflicts* during model merging, TIES-Merging [45] enforces orthogonality constraints, while DARE [49] reduces interference using dropout and rescaling. EMR-Merging [12] aligns task directions and amplitudes through a tuning-free strategy, and MagMax [30] mitigates forgetting through important parameters selection.

Unlike standard TA or the variants mentioned above, which merge entire task vectors without distinction, DuET applies layer-wise Task merging with dynamically calculated retention-adaptation factors to prevent catastrophic forgetting while improving generalization. In Section 6, we benchmark DuET against existing model-merging baselines and demonstrate its superiority across multiple object detectors in the DuIOD task.

3. Methodology

3.1. Problem formulation

In exemplar-free DuIOD, our goal is to develop a unified and robust object detection model that incrementally learns new object classes and adapts to evolving domain characteristics without retaining any past training data (exemplars).

Formally, we define a sequence of tasks as $T = \{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T\}$, where each task $\mathcal{T}_t$ introduces a new set of object classes $\mathcal{C}_t$ from a novel domain $\mathcal{D}_t$. The cumulative set of classes and domains encountered up to task $\mathcal{T}_t$ is denoted by: $\mathcal{C}_{1:t} = \bigcup_{j=1}^t \mathcal{C}_j$ and $\mathcal{D}_{1:t} = \bigcup_{j=1}^t \mathcal{D}_j$, respectively. Furthermore, following the typical CIOD & DIOD assumptions [9, 19, 20, 28, 40, 41], we ensure that:

$$\mathcal{C}_t \cap \mathcal{C}_{1:t-1} = \emptyset, \quad \mathcal{D}_t \cap \mathcal{D}_{1:t-1} = \emptyset, \quad \forall t \geq 2 \quad (1)$$

$$P(\mathcal{D}_t) \neq P(\mathcal{D}_{t'}) \quad \forall t \neq t'. \quad (2)$$

For each task $\mathcal{T}_t$, we define the dataset as $\{\mathcal{X}_t, \mathcal{Y}_t\}$, where $\mathcal{X}_t = \{x_j^t\}_{j=1}^{N_t}$ consists of input images from $\mathcal{D}_t$ that contain only objects from $\mathcal{C}_t$, and $\mathcal{Y}_t = \{y_j^t\}_{j=1}^{N_t}$ provides

bounding box annotations exclusively for $\mathcal{C}_t$. While images in $\mathcal{X}_t$ may include objects from previously learned classes $\mathcal{C}_{1:t-1}$, these objects remain unlabeled and are treated as background. This results in a fundamental *background shift* problem [27, 32], where previously learned objects become part of the background in later tasks—an issue absent in prior works like [35].

To maintain an exemplar-free setting, during task $\mathcal{T}_t$, we have access only to the dataset $\mathcal{D}_t$ and do not retain any images or annotations from previous domains $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_{t-1}\}$. Our objective is to incrementally update the object detection model $\mathcal{M}_{\theta_t}$, parameterized by θ_t , using only data from $\mathcal{D}_t$, ensuring that $\mathcal{M}_{\theta_t}$ can accurately detect objects from all learned classes $\mathcal{C}_{1:t}$ across all encountered domains $\mathcal{D}_{1:t}$.

3.2. Overview of DuET Approach

To tackle DuIOD, we discuss the DuET approach in this and subsequent sections. As depicted in Figure 3, DuET strategically decomposes model parameters into shared (θ_s) and task-specific (θ_{τ_t}) components, thereby facilitating efficient knowledge transfer and modular adaptation. The DuET framework is built upon two core components:

- **DuET Module:** This module fuses shared task vectors (τ_{old} and τ_{curr}) through dynamic, layer-wise weighting factors—retention (α_l) and adaptation (β_l)—as described in Section 3.4. This design not only preserves previously acquired knowledge but also incorporates new information in a balanced manner, effectively countering catastrophic forgetting.
- **Incremental Head:** This component consolidates task-specific parameters from both past and current tasks, thereby enhancing the model’s generalization across diverse detection scenarios. Additionally, **Directional Consistency Loss** ($\mathcal{L}_{\text{DC}}$, see Section 3.5) enforces alignment in the direction of incremental updates and prevents detrimental sign conflicts.

Unlike prior approaches [8, 15, 25], which are optimized for specific detection backbones, DuET exhibits broad generalization across a variety of object detectors [17, 29, 52] (see Table 3). Moreover, DuET’s training and TA algorithm are provided in Algorithms 1 & 2 (suppl.).

3.3. Training Strategy

Base Task: $\mathcal{T}_1$. Consider a generic, pre-trained object detector $\mathcal{M}_{\theta_0}$ parameterized by θ_0 as shown in figure 3. Let Task $\mathcal{T}_1$ introduce new $\mathcal{C}_1$ classes from $\mathcal{D}_1$ domain. We first train this model on the dataset $\{\mathcal{X}_1, \mathcal{Y}_1\}$ for Task $\mathcal{T}_1$ using the standard detection loss, $\mathcal{L}_{\text{Detector}}$, which is specific to the object detector being considered. After the first update step, the model’s parameters become:

$$\theta_1 \leftarrow \theta_0 - \eta \nabla_{\theta} \mathcal{L}_{\text{Detector}}(\theta_0; \mathcal{X}_1, \mathcal{Y}_1), \quad (3)$$

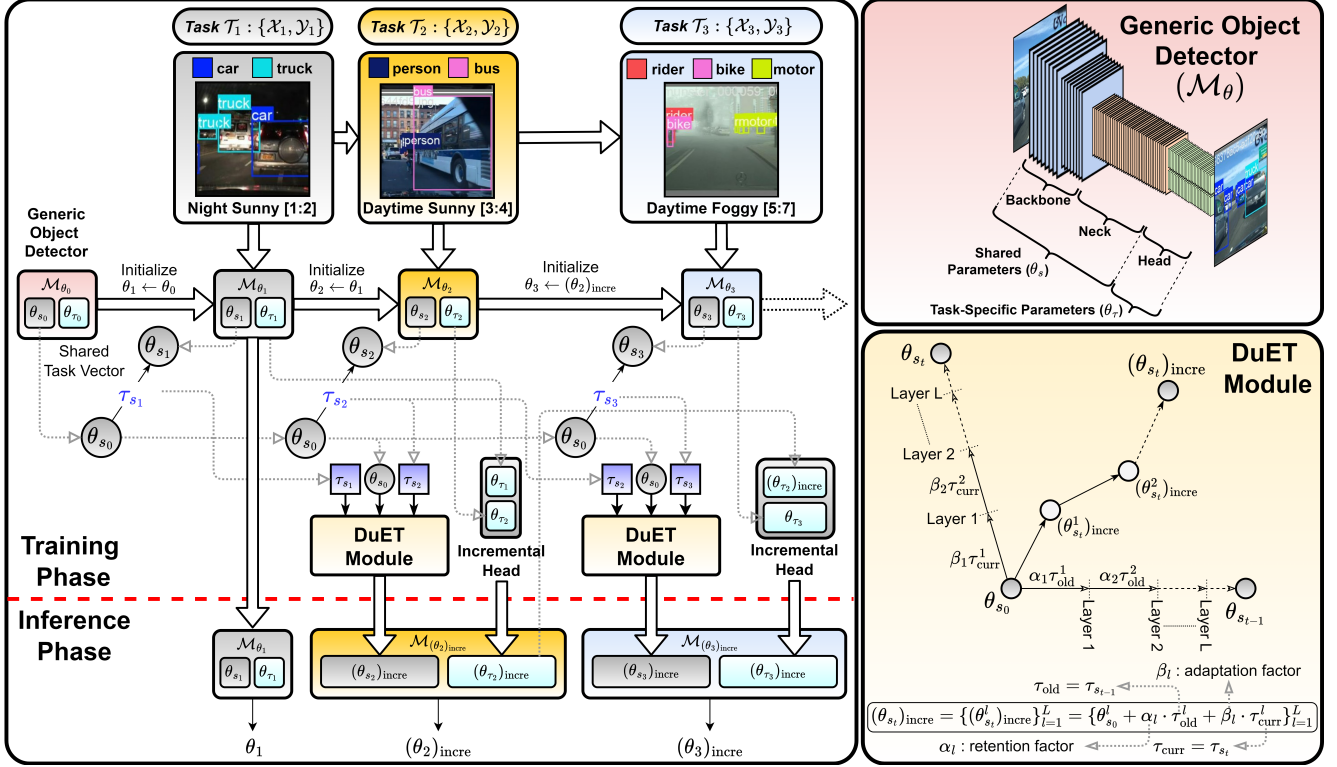


Figure 3. **Overview of the proposed DuET framework for exemplar-free Dual Incremental Object Detection (DuIOD).** **Left:** To setup DuIOD task, we illustrate a sequence of three tasks $\{\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3\}$, each introducing new object classes (e.g. $\{car, truck\}$, $\{person, bus\}$, $\{rider, bike, motor\}$) under different domain shifts (Night Sunny, Daytime Sunny, Daytime Foggy) respectively. A generic object detector $\mathcal{M}_{\theta_0}$ is first fine-tuned on $\mathcal{T}_1$, yielding parameters θ_1 , the parameters of the model at every Task are decomposed into shared (θ_s) and task-specific (θ_{τ_i}) parameters. Subsequently, for each incremental Task ($\mathcal{T}_t, t \geq 2$), the shared parameters are merged via *DuET Module* while the *Incremental Head* concatenates the task-specific parameters from the old and current Task, θ_{τ_t} & $(\theta_{\tau_{t-1}})_{incre}$ respectively. During the *Inference Phase*, the merged incremental weights θ_1 , $(\theta_2)_{incre}$ & $(\theta_3)_{incre}$ are used to detect objects from the respective cumulative sets of classes $\mathcal{C}_1, \mathcal{C}_{1:2}$ & $\mathcal{C}_{1:3}$. **Top-Right:** We depict the decomposition of θ_s & θ_{τ} for a generic, backbone-agnostic object detector $\mathcal{M}_{\theta}$, with the DuET framework validated across a variety of object detectors (e.g. YOLO11 [17], RT-DETR [29]). **Bottom-Right:** We zoom in on the *DuET Module*, which merges the old and current *shared task vectors* (τ_{old}, τ_{curr}) using *layer-wise retention* (α_l) and *adaptation* (β_l) factors as per equation 12 (zoomed-in for best view).

where η is the learning rate.

At every incremental step, we decompose the model parameters into *shared parameters* θ_s , which are common across all incremental tasks, and *task-specific parameters* θ_{τ_t} , which evolve with each incremental step. For instance, in YOLO11 [17], the backbone and neck components, responsible for multi-scale feature extraction and feature refinement, constitute θ_s , whereas the detection head, which is responsible for final task-specific object localization and classification, forms θ_{τ_t} . Hence, for Task $\mathcal{T}_1$, we have:

$$\theta_0 \rightarrow [\theta_{s_0}, \theta_{\tau_0}], \quad \theta_1 \rightarrow [\theta_{s_1}, \theta_{\tau_1}]. \quad (4)$$

After Task $\mathcal{T}_1$ training, following the task vector definition from [13], we compute the first **task vector** as:

$$\tau_{s_1} = \theta_{s_1} - \theta_{s_0}, \quad (5)$$

Here τ_{s_1} effectively captures how the shared parameters must shift to accommodate the first Task, while the remaining parameters, θ_{τ_1} constitute the task-specific weights that

help in localization and classification of the newly introduced $\mathcal{C}_1$ classes. As illustrated in Figure 3, for inference on the base task $\mathcal{T}_1$, we directly employ $\mathcal{M}_{\theta_1}$.

Incremental Tasks $\{\mathcal{T}_2, \mathcal{T}_3 \dots, \mathcal{T}_T\}$. For each incremental task ($\mathcal{T}_t, t \geq 2$), the model encounters a new set of classes $\mathcal{C}_t$ from domain $\mathcal{D}_t$. We follow sequential fine-tuning and initialize $\theta_t \leftarrow \theta_{t-1}$, following [30], but this time, we train the model using $\mathcal{L}_{Total}$ (eq. 3.5), which incorporates a modified Distillation Loss $\mathcal{L}_{Distill}^*$ to help preserve knowledge from previously learned tasks and Directional Consistency Loss $\mathcal{L}_{DC}$ to help resolve sign conflicts and encourage stable parameter updates, along with the object-detector loss $\mathcal{L}_{Detector}$ (see section 3.5). Hence, after every incremental Task $\mathcal{T}_t$, the model parameters are updated as:

$$\theta_t \leftarrow \theta_{t-1} - \eta \nabla_{\theta} \mathcal{L}_{Total}(\theta_{t-1}; \mathcal{X}_t, \mathcal{Y}_t). \quad (6)$$

As before, we then decompose the updated parameters into:

$$\theta_{t-1} \rightarrow [\theta_{s_{t-1}}, \theta_{\tau_{t-1}}], \quad \theta_t \rightarrow [\theta_{s_t}, \theta_{\tau_t}]. \quad (7)$$

Now, to incorporate the knowledge from current domain $\mathcal{D}_t$ and current classes $\mathcal{C}_t$ into the shared parameters θ_s without discarding knowledge of older tasks $\mathcal{T}_{1:t-1}$, we define two **shared task-vectors** as:

$$\tau_{\text{old}} = \theta_{s_{t-1}} - \theta_{s_0}^1, \quad \tau_{\text{curr}} = \theta_{s_t} - \theta_{s_0}^1. \quad (8)$$

The shared task vectors, τ_{old} and τ_{curr} , play a crucial role in capturing the evolution of the shared parameters as the model transitions between tasks. Specifically, τ_{old} represents the cumulative update from the initial pre-trained state up to the last incremental phase, effectively embedding the knowledge of all previous tasks. In contrast, τ_{curr} captures the immediate update introduced by the current Task. This separation allows us to precisely balance the retention of previously learned features with the integration of new information, thereby mitigating catastrophic forgetting while enhancing the model’s generalization capability. By isolating these vectors, the proposed DuET framework benefits from clearly delineating what knowledge needs to be preserved versus what can be adapted.

3.4. Fusing task vectors on shared weights

The core of DuET approach lies in the **DuET Module**, which performs the dynamic integration of τ_{old} and τ_{curr} to update the shared weights (Figure 3). Although these retention-adaptation factors are empirically motivated, they can be viewed through a *stability-plasticity* lens in incremental learning. Interpreting τ_{old}^l and τ_{curr}^l as “directions” in parameter space—one encoding the cumulative shifts for past tasks and the other the shift introduced by the new Task—our layer-wise α_l and β_l define a linear combination that balances preserving old knowledge ($\alpha_l \approx 1$) versus integrating new information ($\beta_l \approx 1$). By comparing $\|\tau_{\text{old}}^l\|$ and $\|\tau_{\text{curr}}^l\|$ at each layer, we obtain a first-order approximation of “which update dominates,” adaptively favoring retention when $\|\tau_{\text{old}}^l\|$ is large and plasticity when $\|\tau_{\text{curr}}^l\|$ is large. Geometrically, this ensures the merged parameters lie neither too close to the old solution nor fully dominated by new updates, effectively mitigating catastrophic forgetting and enhancing generalization. For each layer $l \in \{1, \dots, \mathcal{L}\}$, we define a p-factor:

$$p_l = \frac{\|\tau_{\text{old}}^l\| - \|\tau_{\text{curr}}^l\|}{\|\tau_{\text{old}}^l\| + \|\tau_{\text{curr}}^l\| + \epsilon}, \quad (9)$$

where $\|\cdot\|$ denotes the ℓ_1 norm and ϵ ensures numerical stability. Mapping p_l via a hyperbolic tangent scaled by γ gives:

$$\delta_l = \gamma \tanh(p_l), \quad (10)$$

After clamping δ_l to $[-\gamma, \gamma]$, and adding a base coefficient α_{base} , we define:

$$\alpha_l = \alpha_{\text{base}} + \text{clamp}(\delta_l, -\gamma, \gamma), \quad \beta_l = 1 - \alpha_l. \quad (11)$$

¹Task vectors are computed assuming access to the pre-trained model for all tasks, a standard assumption in task vector-based methods [12, 13].

Hence, unlike second-order methods (e.g., Fisher Merging [31]) that require computing parameter importance matrices, the p-factor captures a simpler, local notion of importance. It forgoes expensive matrix approximations yet still encodes how far each layer’s weights have drifted from the original baseline. Given these per-layer coefficients, the updated shared weights for layer l are:

$$(\theta_{s_t}^l)_{\text{incre}} = \theta_{s_0}^l + \alpha_l \tau_{\text{old}}^l + \beta_l \tau_{\text{curr}}^l. \quad (12)$$

Applying this across all layers yields $(\theta_{s_t})_{\text{incre}}$. Simultaneously, the **Incremental Head** concatenates the current task-specific parameters with the previous ones:

$$(\theta_{\tau_t})_{\text{incre}} \leftarrow [\theta_{\tau_t}; (\theta_{\tau_{t-1}})_{\text{incre}}] \quad (13)$$

Finally, combining these updated shared and task-specific components produces the overall incremental weights:

$$(\theta_t)_{\text{incre}} \leftarrow [(\theta_{s_t})_{\text{incre}}, (\theta_{\tau_t})_{\text{incre}}] \quad (14)$$

which initialize the model for the subsequent Task, i.e., $\theta_{t+1} \leftarrow (\theta_t)_{\text{incre}}$.

As illustrated in Figure 3, for inference on incremental tasks ($t \geq 2$), we employ the merged incremental models $\mathcal{M}_{\theta_t}$, parameterized by the reconstructed weights $(\theta_t)_{\text{incre}}$, to detect $\mathcal{C}_{1:t}$ classes across domains $\mathcal{D}_{1:t}$ following the evaluation protocol detailed in Section 4.2.

3.5. Loss objectives

For the base task ($t = 1$), we optimize the object detector using the standard detection loss, $\mathcal{L}_{\text{Detector}}$, which is specific to the base detector and is responsible for object localization and classification. For incremental tasks ($t \geq 2$), we augment $\mathcal{L}_{\text{Detector}}$ with a modified Distillation Loss $\mathcal{L}_{\text{Distill}}^*$ (discussed in Appendix B), along with our Directional Consistency Loss $\mathcal{L}_{\text{DC}}$ (discussed below) scaled by scaling coefficients λ_{Distill} and λ_{DC} respectively.

Formally, the total loss for task t is:

$$\mathcal{L}_{\text{Total}} = \begin{cases} \mathcal{L}_{\text{Detector}}, & \text{for } t = 1, \\ \mathcal{L}_{\text{Detector}} + \lambda_{\text{Distill}} \mathcal{L}_{\text{Distill}}^* + \lambda_{\text{DC}} \mathcal{L}_{\text{DC}}, & \text{for } t \geq 2. \end{cases} \quad (15)$$

Directional Consistency Loss ($\mathcal{L}_{\text{DC}}$). Incremental learning suffers from conflicting weight updates during model-merging [45], which can destabilize the merging of shared parameters. We introduce the *Directional Consistency Loss* (DC Loss) to address this. This loss penalizes updates in the shared parameter space that diverge in direction relative to previous incremental changes, thus promoting a consistent evolution of the shared task vectors.

Specifically, for $t \geq 2$, we define the DC Loss between consecutive tasks t and $t - 1$ as:

$$\mathcal{L}_{\text{DC}} = \sum_{i \in \theta_s} \text{ReLU} \left[- \left((\tau_{s_t}^{(i)} - \tau_{s_{t-1}}^{(i)}) \cdot (\tau_{s_{t-1}}^{(i)} - \tau_{s_{t-2}}^{(i)}) \right) \right] \quad (16)$$

Here, the dot product quantifies the alignment between successive updates: a negative value indicates a sign conflict (i.e., the current update opposes the previous direction), which the ReLU penalizes. This mechanism enforces a more coherent update trajectory for the shared parameters, effectively mitigating catastrophic forgetting. Consequently, the DC Loss remains object-detector agnostic, as it depends solely on the evolution of shared parameter updates

4. Experimental Details

4.1. Datasets

Following prior works [6, 21, 26, 41, 44], we evaluate DuIOD using two dataset series: *Pascal Series* and *Diverse Weather Series*. *Pascal Series* includes four domains—Pascal VOC [7], Clipart, Watercolor, and Comic [14]. Pascal VOC and Clipart share 20 object categories, while Watercolor and Comic contain six, forming subsets of the former. Similarly, in *Diverse Weather Series*, we consider three diverse weather conditions—Daytime Sunny, Night Sunny, and Daytime Foggy—with seven common object categories, sourced from BDD-100k [48], FoggyCityscapes [5], and Adverse-Weather [11]. The training sequence followed for both dataset series is provided in Appendix A– Table S1, and dataset statistics are detailed in Appendix G.

4.2. Proposed Evaluation Protocol and Metrics

To comprehensively evaluate different methods on the DuIOD task, we propose a new evaluation protocol (see Appendix A– Table S1). Existing metrics such as $\mathcal{F}_{map}$ [4], RSD & RPD [32], and SPMAP [47], primarily assess catastrophic forgetting and do not capture the model’s ability to generalize to unseen classes, which is critical for DuIOD. Since DuIOD goes beyond preserving old knowledge (retention), it also demands evaluating how well the model can adapt to unseen categories (adaptability). We introduce the **Retention-Adaptability Index (RAI)** to address this requirement. RAI is defined as the mean of the Average Retention Index (Avg RI) and Average Generalization Index (Avg GI).

Avg RI quantifies the model’s ability to retain past knowledge and is computed as:

$$\text{Avg RI} = \frac{1}{T-1} \sum_{i=1}^{T-1} RI_{\mathcal{D}_i}, \quad RI_{\mathcal{D}_i} = \frac{\text{mAP}_{\text{old}}^{\mathcal{T}_T}(\mathcal{D}_i[\mathcal{C}_i])}{\text{mAP}_{\text{new}}^{\mathcal{T}_i}(\mathcal{D}_i[\mathcal{C}_i])}. \quad (17)$$

Here, $\text{mAP}_{\text{old}}^{\mathcal{T}_T}$ represents the final-task performance on previously learned classes, while $\text{mAP}_{\text{new}}^{\mathcal{T}_i}$ denotes the initial performance when the classes were first encountered. A higher Avg RI implies better retention of past knowledge while lower values indicate catastrophic forgetting.

Similarly, **Avg GI** quantifies the model’s generalization

to unseen classes and is computed as:

$$\text{Avg GI} = \frac{1}{N} \sum_{(\mathcal{D}_i, \mathcal{T}_j)} GI_{\mathcal{D}_i, \mathcal{T}_j}, \quad GI_{\mathcal{D}_i, \mathcal{T}_j} = \frac{\text{mAP}_{\text{unseen}}^{\mathcal{T}_j}(\mathcal{D}_i[\mathcal{C}_{\text{unseen}}])}{\text{mAP}_{\text{ref}}(\mathcal{D}_i[\mathcal{C}_{\text{unseen}}])}. \quad (18)$$

Here, $\text{mAP}_{\text{unseen}}^{\mathcal{T}_j}$ measures model performance on previously unseen classes, while mAP_{ref} is the reference mAP obtained by training solely on these unseen classes. A higher Avg GI indicates better adaptability to new categories. A detailed discussion, including metric formulations and rationale, is provided in Appendix A.

Moreover, unlike prior works, we do not consider the upper-bound joint-training baseline, as Avg RI and Avg GI compute relative performance retention and generalization through direct ratio-based comparisons. This ensures that the overall metric (RAI) directly reflects catastrophic forgetting without biases introduced by architecture-specific joint-training performance.

5. Results & Discussion

5.1. Quantitative Results

To evaluate the effectiveness of the DuET, we compare it against five baselines: Sequential Fine-Tuning, LwF [25] and ERD [8] (common IOD methods), LDB [41] and CL-DETR [28] (**current open-source SOTA** in DIOD and CIOD). We re-implemented all under the proposed DuIOD setting, reporting the best results across seven experiments—five two-phase and two multi-phase. Results from two two-phase and two multi-phase experiments are in Table 1 and Table 2, while comprehensive results from all experiments are provided in Appendix E. Moreover, as detailed in Appendix C3, DuET maintains a consistent performance irrespective of the randomness in class-domain orders.

We begin with naïve **Sequential Fine-Tuning** using YOLO11n [17] as the base detector, where we observe severe catastrophic forgetting, with Avg RI < 1% across all experiments. **LwF** and **ERD**, both implemented on YOLO11n by modifying the Ultralytics pipeline [17] to incorporate distillation losses (LwF) and the Elastic Response Selection algorithm (ERD), show moderate RAI improvement but gives poor results in multi-phase experiments (Table 2) compared to DuET. On the DIOD side, **LDB**² with VitDet [24] as the base detector suffers from severe catastrophic forgetting, particularly in cross-domain uneven class shifts (Table 1). On the CIOD side, **CL-DETR**² with Deformable DETR [52] struggles under severe domain shifts despite having more trainable parameters. In contrast, DuET consistently outperforms all methods despite having significantly fewer trainable parameters (Table 1) while also demonstrating strong generalization across multiple object detection architectures (Table 3).

DuET achieves an RAI improvement of **+9.82%** and **+13.12%** while preserving **87.44%** and **89.30%** Avg RI

Table 1. Detailed results of various methods on Daytime Sunny [1:4] $\rightarrow$ Night Sunny [5:7], where the expanded columns ($\mathcal{T}_1$: Daytime Sunny [1:4], $\mathcal{T}_2$: Night Sunny [5:7]) show mAP@0.5%, best among the columns in **bold**, second best underlined.

Method	Base Detector	Trainable Params (M)	T1	T2: Night Sunny [5:7]				Avg RI (%)	Avg GI (%)	RAI (%)
			Daytime Sunny [1:4]	Old	New	Unseen				
				Daytime Sunny [1:4]	Night Sunny [5:7]	Night Sunny [1:4]	Daytime Sunny [5:7]			
Sequential FT	YOLO11n	2.58	49.40±0.3	0.00±0.0	62.20±0.5	12.60±0.4	35.90±0.3	0.00±0.0	45.88±0.6	22.94±0.3
LwF _{ECCV'16} [25]	YOLO11n	2.58	49.40±0.2	27.60±0.4	0.34±0.6	21.30±0.3	0.67±0.5	55.87±0.3	21.88±0.7	38.88±0.6
ERD _{CVPR'22} [8]	YOLO11n	2.58	49.40±0.5	33.00±0.4	34.00±0.3	26.10±0.6	29.10±0.7	66.80±0.5	53.04±0.3	59.92±0.4
LDB _{AAAI'24} [41]	VitDet	110.52	45.30±0.6	0.50±0.3	15.10±0.4	0.30±0.5	16.90±0.7	1.10±0.2	22.41±0.3	11.76±0.6
CL – DETR _{CVPR'23} [28]	Deformable DETR	39.85	46.29±0.4	27.41±0.5	31.94±0.6	19.85±0.3	32.55±0.4	59.21±0.2	54.96±0.5	57.09±0.4
DuET (Ours)	YOLO11n	2.58	49.40±0.2	43.50±0.1	22.20±0.3	31.60±0.2	27.40±0.1	88.06±0.2	56.95±0.1	72.51±0.2

Table 2. Results of two-phase and multi-phase experiments on DuIOD task. Best among the columns in **bold**, second best underlined.

Method	Base Detector	VOC [1:10] $\rightarrow$ Clipart [11:20]			Watercolor [1:3] $\rightarrow$ Comic [4:6] $\rightarrow$ Clipart [7:13] $\rightarrow$ VOC [14:20]			Night Sunny [1:2] $\rightarrow$ Daytime Sunny [3:4] $\rightarrow$ Daytime Foggy [5:7]		
		Avg RI (%)	Avg GI (%)	RAI (%)	Avg RI (%)	Avg GI (%)	RAI (%)	Avg RI (%)	Avg GI (%)	RAI (%)
Sequential FT	YOLO11n	0.75 $\pm$ 0.2	12.86 $\pm$ 0.4	6.81 $\pm$ 0.3	0.00 $\pm$ 0.0	11.05 $\pm$ 0.5	5.53 $\pm$ 0.3	0.00 $\pm$ 0.0	30.51 $\pm$ 0.6	15.26 $\pm$ 0.5
LwF [25]	YOLO11n	<u>72.64</u> $\pm$ 0.3	33.74 $\pm$ 0.5	53.19 $\pm$ 0.4	52.66 $\pm$ 0.6	17.01 $\pm$ 0.4	34.84 $\pm$ 0.3	27.94 $\pm$ 0.7	23.78 $\pm$ 0.5	25.86 $\pm$ 0.6
ERD [8]	YOLO11n	68.66 $\pm$ 0.4	<u>43.68</u> $\pm$ 0.3	<u>56.17</u> $\pm$ 0.6	54.76 $\pm$ 0.5	<u>41.13</u> $\pm$ 0.4	47.95 $\pm$ 0.7	44.60 $\pm$ 0.6	39.40 $\pm$ 0.5	42.00 $\pm$ 0.3
LDB ² [41]	VitDet	66.89 $\pm$ 0.5	18.76 $\pm$ 0.3	42.83 $\pm$ 0.4	<u>86.08</u> $\pm$ 0.6	19.57 $\pm$ 0.5	52.83 $\pm$ 0.4	50.50 $\pm$ 0.6	5.42 $\pm$ 0.7	27.96 $\pm$ 0.5
CL-DETR ² [28]	Deformable DETR	68.29 $\pm$ 0.3	40.72 $\pm$ 0.4	54.51 $\pm$ 0.3	71.73 $\pm$ 0.5	36.63 $\pm$ 0.6	<u>54.18</u> $\pm$ 0.4	<u>64.26</u> $\pm$ 0.3	43.46 $\pm$ 0.5	<u>53.86</u> $\pm$ 0.6
DuET (Ours)	YOLO11n	87.44 $\pm$ 0.2	44.54 $\pm$ 0.1	65.99 $\pm$ 0.3	89.30 $\pm$ 0.2	42.60 $\pm$ 0.1	65.95 $\pm$ 0.2	88.57 $\pm$ 0.2	<u>41.92</u> $\pm$ 0.1	65.25 $\pm$ 0.2

Table 3. Results of proposed DuET framework on Daytime Sunny [1:4] $\rightarrow$ Night Sunny [5:7] experiment with different base detectors. Best among the columns in **bold**, second best underlined.

Method	Base Detector	Trainable Params (M)	GFLOPs	Avg RI (%)	Avg GI (%)	RAI (%)
DuET (Ours)	VitDet [24]	110.52	1829.61	27.55 $\pm$ 0.3	28.22 $\pm$ 0.8	27.89 $\pm$ 0.5
	Deformable DETR [52]	39.85	<u>11.77</u>	84.45 $\pm$ 0.2	33.45 $\pm$ 0.1	58.95 $\pm$ 0.2
	RT-DETR-l [29]	<u>32.00</u>	103.4	47.73 $\pm$ 0.2	21.00 $\pm$ 0.1	34.37 $\pm$ 0.2
	RT-DETR-x [29]	65.49	222.5	56.39 $\pm$ 0.2	24.15 $\pm$ 0.1	40.27 $\pm$ 0.2
	YOLO11n [17]	2.58	6.3	<u>88.06</u> $\pm$ 0.2	56.95 $\pm$ 0.1	72.51 $\pm$ 0.2
	YOLO11x [17]	56.84	194.4	96.88 $\pm$ 0.2	<u>42.41</u> $\pm$ 0.1	<u>69.18</u> $\pm$ 0.2

Table 4. Ablations of DuET framework with different components and losses on VOC [1:10] $\rightarrow$ Clipart [11:20] with YOLO11n as base detector.

Seq FT	Incremental Head	DuET Module	$\mathcal{L}_{\text{Distill}}^*$	$\mathcal{L}_{\text{DC}}$	Avg RI (%)	Avg GI (%)	RAI (%)
$\times$	$\times$	$\times$	$\times$	$\times$	0.5	9.13	4.82
$\checkmark$	$\times$	$\times$	$\times$	$\times$	0.75	12.86	6.81
$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	24.75	33.36	29.06
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$	75.00	37.26	56.13
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	87.06	37.75	62.41
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	87.44	44.54	65.99

on the *Pascal Series* two-phase and multi-phase experiments, respectively. Similarly, on the Diverse Weather Series, DuET achieves an RAI improvement of **+12.59%** and **+11.39%** while preserving **88.06%** and **88.57%** Avg RI on the two-phase and multi-phase experiments, respectively.

5.2. Qualitative Analysis

In Figure 4, we present qualitative comparisons of different methods on two-phase experiments, while additional qualitative results on multi-phase experiments are provided in

²CL-DETR [28] relies on a DETR-based framework with bipartite matching and cannot be trivially adapted to YOLO11. LDB [41], designed for learning domain biases, was unable to handle simultaneous class-domain shifts of DuIOD with YOLO11 (often yielding 0 mAP and at least a 40% drop in RAI compared to its original backbone), so we compare each method on its own backbone.

Appendix E2. In *VOC [1:10] $\rightarrow$ Clipart [11:20]*, LwF and LDB misclassify the `cow` class (introduced in $\mathcal{T}_1$: VOC [1:10]) as `horse` or `sheep` (both introduced in $\mathcal{T}_2$: Clipart [11:20]). While ERD and CL-DETR correctly classify it, they exhibit significantly lower confidence scores compared to DuET. A similar pattern is observed for the `person` and `motorbike` classes, both introduced in $\mathcal{T}_2$, where only DuET correctly detects both objects in the unseen VOC [11:20], while competing methods mostly fail.

Similarly, in *Daytime Sunny [1:4] $\rightarrow$ Night Sunny [5:7]*, the `car` class (introduced in $\mathcal{T}_1$: Daytime Sunny [1:4]) undergoes a severe domain shift when evaluated in unseen Night Sunny [1:4]. As a result, most methods fail to detect a majority of `car` instances, instead misclassifying them as `background`, because of the background shift challenge in DuIOD. (see Appendix F for more details). However, DuET effectively overcomes this challenge by correctly detecting most of the `car` instances (improving mAP_{car}^{T₂} from **49.4** to **71.8**). Similar misclassifications are observed for the `truck` class, introduced in $\mathcal{T}_2$: Night Sunny [5:7], where most methods struggle to detect it in unseen Daytime Sunny [5:7], while DuET provides consistent detections. These qualitative results highlight the effectiveness of DuET in seamlessly adapting to unseen classes in new domains.

6. Main Ablation Studies

This section covers the main ablation studies, while extended ablations, including the **impact of loss components**, **hyperparameter sensitivity analysis**, the **influence of random class-domain order**, and **complexity analysis**, are provided in Appendix C.

Role of Sequential Fine-tuning. While DuIOD inherently involves sequential learning, we compare independent



Figure 4. **Qualitative comparisons on two-phase experiments for different methods on the DuIOD task.** The *top two rows* display detection results on unseen classes: Clipart [1:10] & VOC [11:20], for: VOC [1:10] $\rightarrow$ Clipart [11:20] experiment. Similarly, the *bottom two rows* show detection results on unseen classes: Night Sunny[1:4] & Daytime Sunny[5:7], for: Daytime Sunny[1:4] $\rightarrow$ Night Sunny[5:7] experiment (zoomed-in for best view).

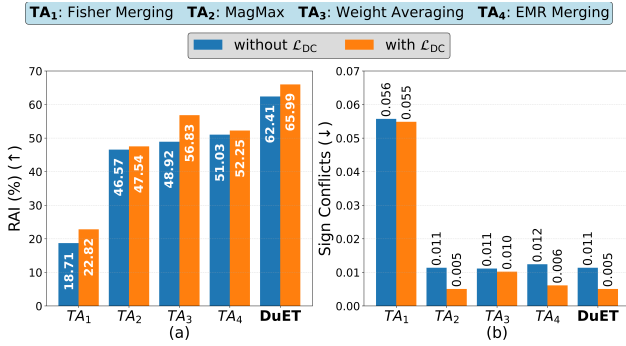


Figure 5. Impact of $\mathcal{L}_{DC}$ on (a) improving RAI and (b) reducing sign conflicts across different TA approaches, evaluated on VOC [1:10] $\rightarrow$ Clipart [11:20] using YOLO11n with Incremental Head and sequential fine-tuning.

(first row, Table 4) vs. sequential fine-tuning (second row) in DuET framework, observing a significant improvement in Avg GI and, hence, RAI with the latter, aligning with the findings in [30], highlighting its importance in DuIOD.

Effectiveness of DuET Module and Incremental Head. Next, we evaluate the effectiveness of the DuET Module and Incremental Head in the DuET framework. Incremental Head (third row) brings a substantial improvement in both Avg RI and Avg GI over Seq FT. This highlights the importance of concatenating task-specific parameters from previous and current tasks, thereby enhancing the model’s generalization capability. Incorporating the DuET Module further leads to a significant improvement in both Avg RI and Avg GI (fourth row), effectively mitigating catastrophic forgetting while enhancing generalization.

Impact of $\mathcal{L}_{Distill}^*$ and $\mathcal{L}_{DC}$. We further investigate the impact of $\mathcal{L}_{Distill}^*$ and $\mathcal{L}_{DC}$ in the proposed DuET approach.

As shown in Table 4 (rows five and six), both losses contribute significantly. While $\mathcal{L}_{Distill}^*$ brings a substantial improvement in Avg RI by helping to retain knowledge from previous tasks, $\mathcal{L}_{DC}$ further enhances Avg GI, thereby improving the model’s generalization capability. The combination of both losses yields the best performance, achieving the highest Avg RI and Avg GI, highlighting their critical role in the DuET approach.

Comparison with other model-merging algorithms. In Figure 5, we compare the performance of related TA approaches with the proposed approach. The results indicate that $\mathcal{L}_{DC}$ improves RAI by an average of **9.67%** while significantly reducing sign conflicts by an average of **34%**, averaged across all TA approaches, with DuET achieving the best overall performance.

7. Conclusions

In this paper, we introduced DuET, a task-arithmetic-driven solution for robust IOD, enabling models to learn new object categories while generalizing across evolving domains incrementally. Unlike previous works that focus on either class or domain shifts, DuET integrates both, making it practical for real-world applications like autonomous driving and surveillance. We demonstrated its effectiveness across diverse object detectors like YOLO11 & RT-DETR through extensive experiments on the *Pascal Series* and *Diverse Weather Series* datasets. Additionally, we proposed Directional Consistency Loss to mitigate sign conflicts and a Retention-Adaptability Index to comprehensively evaluate retention and adaptation. DuET opens a new research direction in IOD, providing a promising framework for continual learning and domain adaptation.

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